MIL-W-47206(MI)
12 July 1974
SUPERSEDING
MIS-14296B
9 April 1970

MILITARY SPECIFICATION

WIRE AND CABLE, ELECTRICAL, SINGLE CONDUCTOR. TWISTED PAIRS AND MULTICONDUCTOR SHIELDED AND JACKETED, HIGH TEMPERATURE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for electrical wire in twisted pairs and multiconductor shielded and jacketed cable suitable for use in the temperature range from minus 65 to plus 200 degrees Celsius (C) (minus 85 to plus 392 degrees Fahrenheit (F)).

1.2 Classification.

1.2.1 Cables. The cables covered by this specification shall be furnished according to the following classes as characterized by wire type, braided copper wire shield strand coating and outer jacket insulation.

FSC 6145

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Cable Class (3.2.1)	Wire Type (3.2.5)	Shield Coating (3.2.6)	Outer Jacket Insulation
I	B	Tin	Polyvinylchloride (PVC)
II	E	Silver	Polytetrafluoroethylene (TFE)

1.2.2 Twisted pairs. The twisted pairs of wires covered by this specification shall be furnished according to the following classes as characterized by wire type.

Twisted Pair Class (3.2.3)	Wire Type (3.2.5)
III	B
IV	E

1.2.3 Single conductor. The single conductor wire covered by this specification shall be furnished according to the following class as characterized by the wire type.

Single Conductor Class (3.2.7)	Wire Type (3.2.5)
V	E

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-W-343	Wire, Electrical and Non-electrical, Copper, Uninsulated

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Military

MIL-W-16878	Wire, Electrical, Insulated High Temperature
MIL-W-16878/1	Wire, Electrical, Type B, 105°C, 600 Volts, (Insulated, High Temperature)
MIL-W-16878/4	Wire, Electrical, Type E, 200°C and 260° C, 600 Volts, (Insulated, High Temperature)
MIL-C-27500	Cable, Electrical, Shielded and Unshielded, Aircraft and Missile

STANDARDS

Federal

FED-STD-228	Cable and Wire, Insulated, Methods of Testing

Military

MIL-STD-104	Limits for Electrical Insulation Color
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-681	Identification Coding and Application of Hookup and Lead Wire

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM B 298	Silver-Coated Soft or Annealed Copper Wire

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample (see 4.2) of material shall meet the requirements of this specification.

3.2 Materials.

3.2.1 Class I construction. Class I cables shall consist of two, three or four Type B wires of the same size (see 3.2.6) twisted together without fillers to form the cable core. The cable core shall be encased in tin coated copper shielding braid (see 3.2.7) and covered by an extruded PVC outer jacket insulation.

3.2.2 Class II construction. Class II cables shall consist of two, three or four Type E wires of the same size (see 3.2.6) twisted together without fillers to form the cable core. The cable core shall be encased in silver coated copper shielding braid (see 3.2.7) and covered by an extruded solid or laminated virgin TFE outer jacket insulation.

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3.2.3 Class III construction. Class III twisted pairs shall consist of two Type B wires of the same size twisted together without shielding or outer jacket insulation.

3.2.4 Class IV construction. Class IV twisted pairs shall consist of two Type E wires of the same size twisted together without shielding or outer jacket insulation.

3.2.5 Class V construction. Class V single conductor shall consist of one Type E wire encased in silver coated copper shielding braid (see 3.2.7) and covered by and extruded solid or laminated virgin TFE outer jacket insulation.

3.2.6 Wire. The wire used for single conductor, twisted pairs and cables specified herein shall conform to MIL-W-16878/1 for Type B and MIL-W-16878/4 for Type E, modified respectively for sizes 18 AWG through 32 AWG to have a minimum of 19 strands. The Type B wire shall have a polyamide jacket extruded over the PVC primary insulation in conformance with MIL-W-16878. The wire conductors shall conform to the nominal American Wire Gauge (AWG) sizes as specified by the procuring activity (see 3.5).

3.2.6.1 Color coding. Color coding of wire for single conductor, twisted pairs and cables shall be in accordance with MIL-STD-681, as specified (see 3.5).

3.2.7 Shielding. The cable shielding material shall be manufactured from copper wire or strands conforming to QQ-W-343. Prior to braiding, the material shall be tin coated with commercially pure tin or silver coated with commercially pure silver as required by the applicable cable class (see 1.2.1). The silver coating shall meet the thickness requirement specified in 3.4.2.

3.3 Electrical properties.

3.3.1 Outer jacket insulation. The outer jacket insulation of cables shall be capable of withstanding 2000 volts alternating current (vac) at a frequency of 60 Hertz (Hz) at ambient temperature without evidence of sparking or breakdown.

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3.3.2 Dielectric strength (dry). The cables shall show no evidence of breakdown between the conductors and braided shield when subjected to 1500 vac at a frequency of 60 Hz at ambient temperature.

3.3.3 Dielectric strength (wet). The cables shall show no evidence of breakdown between the braided shield and outer jacket insulation when submerged in water and subjected to 1000 vac at a frequency of 60 Hz.

3.4 Physical properties.

3.4.1 Length of lay of wires. The length of lay of wires in cables and twisted pairs shall be 8 to 16 times the outside diameter of the cable.

3.4.2 Silver coating thickness. The silver coating thickness of the braided shielding material shall be not less than 40 microinches.

3.4.3 Tin and silver coating continuity. The tin and silver coating of the braided shielding shall be continuous and shall show no evidence of blackening.

3.4.4 Shield coverage. The shield coverage shall be not less than 85 percent.

3.4.5 Shield braid angle. The angle of the shield braid shall be 35 plus 5, minus 10 degrees.

3.4.6 Outer jacket insulation thickness. The outer jacket insulation thickness shall be not less than 0.010 inch.

3.4.7 Outer jacket color. The outer jacket color shall be white in accordance with MIL-STD-104, Class 1.

3.4.8 Cable diameter. The outside diameter (OD) of the finished cables shall conform to the dimensions specified in Table I.

3.4.9 Heat resistance. The outer jacket insulation of the cables shall show no evidence of cracking after exposure to 136 degrees C (276.8 degrees F) of Class I cables and 250 degrees C (482 degrees F) of Class II cables.

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Table I

Cable Diameter

AWG wire size	Number of wires	Finished cable OD, ± .010 inch	
		Class I	Class II
26	2	0.133	0.126
24	2	0.148	0.138
22	2	0.163	0.150
20	2	0.183	0.164
18	2	0.208	0.188
16	2	0.233	0.215
14	2	0.260	0.241
12	2	-	0.279
26	3	0.141	0.136
24	3	0.156	0.145
22	3	0.173	0.158
20	3	0.194	0.175
18	3	0.222	0.199
16	3	0.248	0.225
14	3	0.269	0.255
12	3	-	0.296
26	4	0.155	0.145

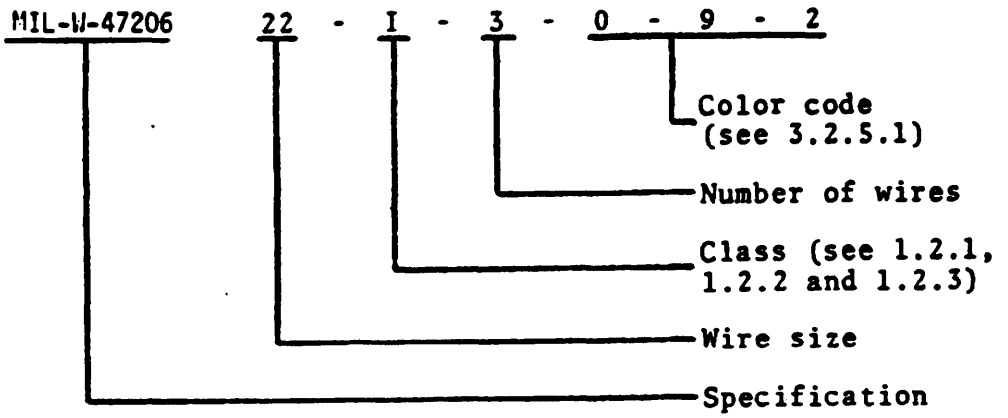
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Table I (Continued)

24	4	0.173	0.159
22	4	0.190	0.176
20	4	0.215	0.200
18	4	0.245	0.230

3.4.10 Cold resistance. The outer jacket insulation of cables shall show no evidence of cracking after exposure to minus 65 degrees C (minus 85 degrees F).

3.5 Drawing callout. Cables and twisted pairs shall be specified on the drawing in accordance with the following example. The drawing callout information shall be considered as the identifying part number



3.6 Workmanship. The workmanship shall be such as to insure a product which is uniform and in conformance with the requirements of this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of material. Preproduction samples which do not meet all the requirements of this specification shall be subject to rejection and returned to the supplier.

4.3 Classification of examinations and tests. The examination and testing of the single conductor, twisted pairs and cables shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.4 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.5 Quality conformance tests.

4.5.1 Single conductor, twisted pairs and cables. Quality conformance tests for acceptance shall consist of the following examinations and tests:

a. Dielectric strength (dry).

b. Dielectric strength (wet).

c. Length of lay of wires (where applicable).

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d. Tin and silver coating continuity.

e. Shield coverage.

f. Shield braid angle.

g. Outer jacket insulation thickness.

h. Color coding of wires.

i. Heat resistance.

j. Cold resistance.

4.5.2 Twisted pairs and cables. Quality conformance tests for acceptance of twisted pairs and cables shall conform to the applicable tests in MIL-W-16878/1 and MIL-W-16878/4 for the wires and length of lay of wires (see 4.6.2.1).

4.5.3 Lot size and sampling.

4.5.3.1 Lot size. Lot size shall consist of all the single conductor, twisted pairs or cables submitted for acceptance at the same time; which have been prepared by the same company without change in materials or processes in one continuous period of operation.

4.5.3.2 Sampling. Not less than three 5-foot sample lengths of single conductor, twisted pairs and cables shall be selected at random from each lot for all quality conformance tests except for the dielectric strength (dry) test (see 4.6.1.2). Sampling for the dielectric strength (dry) test shall be 100 percent of production. Failure of sample to pass the quality conformance tests shall be cause for lot rejection.

4.6 Test methods.

4.6.1 Electrical property tests.

4.6.1.1 Outer jacket insulation. The cable shall be tested in accordance with FED-STD-228, Method 6211. The test

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voltage shall be 2000 vac at 60 hz. The cable shall show no evidence of sparking or breakdown. The supplier shall furnish certification with each lot of material stating compliance with the requirement.

4.6.1.2 Dielectric strength (dry). The cable shall be tested for dielectric strength (dry) by applying 1500 vac at 60 Hz between each conductor of the multiconductor cable and other conductors electrically connected together with the shield braid. The test voltage shall be maintained for not less than 5 seconds at each application. The cable shall show no evidence of sparking or breakdown.

4.6.1.3 Dielectric strength (wet). The cable shall be tested for dielectric strength (wet) by immersing the cable specimen in a water bath at ambient temperature for not less than 1 hour and then testing while still submerged at 1000 vac at 60 Hz for not less than 5 seconds.

4.6.2 Physical property tests.

4.6.2.1 Length of lay of wires. The length of lay of wires in the cables and twisted pairs shall be measured. The length of lay of wire shall be 8 to 16 times the diameter of the cable.

4.6.2.2 Silver coating thickness. The silver coating thickness of the braided shielding material shall be determined in accordance with ASTM B 298, electric determination method. The silver coating thickness shall be not less than 40 microinches.

4.6.2.3 Tin and silver coating continuity. The tin and silver coating continuity shall be determined in accordance with FED-STD-228, Method 7121. The coating shall continuous and shall show no evidence of blackening.

4.6.2.4 Shield coverage. The shield coverage shall be determined in accordance with FED-STD-228, Method 8121. The shield coverage shall be not less than 85 percent.

4.6.2.5 Shield braid angle. The shield braid angle shall be determined in accordance with FED-STD-228, Method 1631. The angle of the shield braid shall be 35 plus 5, minus 10 degrees.

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4.6.2.6 Outer jacket insulation thickness. The outer jacekt insulation thickness shall be determined in accordance with FED-STD-228, Method 1018. The outer jacket insulation thickness shall be not less than 0.010 inch.

4.6.2.7 Outer jacket color. The outer jacket color shall be visually examined for compliance with MIL-STD-104, color white, Class 1.

4.6.2.8 Cable diameter. Cable diameters shall be measured and shall be as specified in Table I.

4.6.2.9 Color coding of wires. Color coding shall be visually examined and shall conform to MIL-STD-681.

4.6.2.10 Heat resistance. Class I and Class II cable specimens shall be placed into a forced draft air oven and subjected to temperature aging for not less than 96 hours at the specified temperatures (see 3.4.9). The specimens shall then be removed from the oven, allowed to cool at ambient temperature for 30 to 40 minutes and then wrapped around a mandrel for not less than 5 close turns. The mandrel diameter shall be 3 to 4 times the OD of the finished cable. The specimens shall be removed from the mandrel as a helical coil and visually examined for cracks in outer jacket insulation without the aid of magnification. The cable specimens shall then be subjected to the test of 4.6.1.3.

4.6.2.11 Cold resistance. The cable specimens shall be tested in accordance with MIL-C-27500 to determine compliance with the cold resistance requirement (see 3.4.10). After completion of the cold resistance (cold bend) test, the specimens shall be subjected to the test of 4.6.1.3.

4.7 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be examined to determine compliance with Section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified in the purchase order, the single conductor, twisted pairs and cables shall be packaged on reels or on spools, each having an appropriate diameter for the respective size. In no case shall the eye of the coil, the barrel, the reel, or spool be less than 4 inches or 14 times the diameter of the finished single conductor, twisted pairs or cable. Unless otherwise specified in the purchase order (see 6.2), the material shall be supplied in lengths not less than 100 feet. The material shall be prepared for shipment in such a manner that damage during shipment is prevented.

5.2 Marking. In addition to any special marking required by the contract or purchase order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.2.1 Special marking. Unless otherwise specified in the purchase order, each reel, spool, or coil of single conductor, twisted pairs and cable shall be identified with the part number (see 3.5).

6. NOTES

6.1 Intended use. The single conductors, twisted pairs and cables are intended to be used for internal hook-up of general missile and associated electrical and electronic equipment.

6.2 Ordering data. Porcurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required and, if so, pertinent details (see 3.1).

c. Sampling plan if other than specified (see 4.5.3.2).

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d. Length of material on reels or spools if other than specified (see 5.1).

e. Material identifying part number (see 5.2.1).

f. Any special marking requirements if other than specified (see 5.2.1).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS 14296, dated 9 April 1970.

Custodian:
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Preparing Activity:
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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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